Hyungjun Kim

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LinkedIn · GitHub · Hugging Face

SUMMARY

Undergraduate student at Seoul National University majoring in Civil and Environmental Engineering with an interdisciplinary major in Artificial Intelligence. Strong interest in Large Language Models (LLMs), especially quantization, efficient inference, and deploying AI systems to real-world industrial environments. Hands-on experience through internships at AI startups, research labs, and production engineering teams.

EXPERIENCE

Solution Architect Intern 2026.03 – Present
SqueezeBits

- Worked on hardware-aware software optimization for efficient model inference
- Collaborated with Modular (Mojo / MAX framework) to accelerate diffusion model inference
- Investigated FP8 quantization slowdowns in FLUX-based diffusion models
- Profiled GPU kernel dispatch efficiency, memory movement, and scaling overhead
- Adapted rapidly to large production codebases with frequent upstream changes

ML Research Engineer Intern 2026.01 – 2026.02
AsteroMorph

- Optimized repeated inference and weight-update loops using vLLM
- Tuned FP8 kernels for efficient execution on NVIDIA B200 GPUs
- Improved RAG retrieval performance and trained reranker models
- Built automated synthetic data pipelines using LLMs
- Designed multi-LLM ensemble ranking systems for evaluation workflows

Undergraduate Research Intern 2025.06 – 2025.12
Seoul National University MLLab

- Conducted research on LLM quantization for efficient inference
- Investigated performance gaps between theoretical and real GPU speedups
- Contributed to FlatQuant open-source repository
- Studied research methodology and paper writing for top conferences

Undergraduate Research Intern 2024.10 – 2025.06
Seoul National University CMALab

- Studied GPTQ, AWQ, QuaRot and related quantization methods
- Implemented research ideas using PyTorch and Transformers
- Assisted research on reasoning-oriented LLM optimization
- Developed collaborative research experience in academic environments

EDUCATION

B.S. Civil and Environmental Engineering / I.M. Artificial Intelligence 2020.03 – Present
Seoul National University GPA: 3.92

SKILLS

Languages: Python, C++, SQL, Bash, MATLAB
AI / Deep Learning: PyTorch, Transformers, LoRA, Fine-tuning, Quantization
LLM Systems: vLLM, CUDA, TensorRT-LLM, Efficient Inference, Prompt Engineering
Backend / Infra: Linux, Git, Docker, FastAPI, REST API

PROJECTS

K-EXAONE Quantization — Quantized LG AI Research K-EXAONE model into W4A16 AWQ format

- Published deployable checkpoints on Hugging Face
- Validated compatibility with vLLM serving stack
- Benchmarked latency and throughput against competing models

FlatQuant Optimization — Performance analysis and optimization of FlatQuant post-training quantization

- Investigated why theoretical quantization speedups were not achieved in real GPU inference
- Analyzed kernel bottlenecks, memory movement, and dispatch overhead
- Contributed fixes and improvements to open-source repository

H-Trust — AI trust decision support platform powered by Gemini API

- Built modular reasoning pipeline and deployable web architecture
- Used AI coding agents for planning, coding, debugging, and refactoring
- Won 1st place in Hanwha Life Future Talent Contest

MKCV — YAML-driven CV / resume generation platform built with AI coding agents

- Designed a single-source-of-truth resume system using YAML schemas
- Leveraged Claude Code agent workflows for rapid planning, frontend/backend coding, refactoring, and iteration
- This CV was also generated using the MKCV platform

ARC Reasoning LLM — Lightweight Abstract Reasoning Corpus solver using quantized Qwen LLM

- Used 4-bit quantization and LoRA fine-tuning on 0.6B model
- Proposed shape-grid decoupled custom loss for better generalization
- Applied geometric augmentation and ensemble decoding

Pacman Game — Full C++ Pac-Man clone with gameplay systems and AI ghost logic

- Implemented maze navigation, collisions, score system, and life resets
- Built target-based movement logic for multiple ghost characters
- Delivered fully playable end-to-end desktop game

AWARDS

1st Place, Hanwha Life Future Talent Contest - AI sector 2026
Hanwha Life

Excellence Award, 40th SNU Liberal Education Report Competition 2024
Seoul National University

Special Award, Future Marine Science and Technology Talent Award 2024
KAOSTS

Silver Prize, 32th Korea Mathematical Olympiad 2018
KMO

Gold Prize, 30th Korea Junior Mathematical Olympiad 2016
KMO

EXTRACURRICULAR ACTIVITIES

Reporter / Vice President 2020-2024
Gongsang, SNU College of Engineering

- Wrote official promotional articles for College of Engineering
- Mentored students in outreach engineering programs
- Developed leadership, writing, and communication skills